

RWE / Advanced Analytics 实习简历条目（可直接复制）

针对 Real World Evidence & Analytics Summer Intern JD 优化

项目经历 | 英文版

MIMIC-IV ICU Real-World Evidence Study: Early Vasopressor Initiation and In-Hospital Mortality | Python, SQL/DuckDB

- Designed a target-trial-emulation style observational RWE study using MIMIC-IV ICU EHR data, defining adult sepsis-coded first ICU stays, 6-hour landmark/grace period, early vasopressor exposure, hospital mortality outcome, and baseline confounders.
- Built a patient-level analysis dataset by integrating ICU stays, admissions, diagnoses, medication/input events, vitals, labs, and mortality records; constructed an analytic cohort of 9,502 ICU stays with reproducible cohort flow and treatment classification.
- Conducted propensity-score diagnostics and causal adjustment using stabilized IPTW, weighted logistic regression, and AIPW; generated Table 1/SMD balance assessment, propensity score overlap plots, love plots, and adjusted risk-difference estimates.
- Compared crude and adjusted findings: early vasopressor users had higher crude mortality (32.9% vs. 25.2%), while AIPW-adjusted risk difference was close to null (1.17 percentage points; 95% CI: -0.52 to 2.87), illustrating confounding by indication in ICU treatment strategy analysis.

项目经历 | 中文版

MIMIC-IV ICU 真实世界证据研究：早期升压药使用与院内死亡风险 | Python, SQL/DuckDB

- 基于 MIMIC-IV ICU 真实世界 EHR 数据，设计 target trial emulation 风格的观察性研究，定义成人 sepsis-coded 首次 ICU 患者、6 小时 landmark/grace period、早期升压药暴露、院内死亡结局和 baseline confounders。
- 使用 SQL/DuckDB 与 Python 整合 ICU stay、admission、diagnosis、input events/medication、vitals、labs 和 mortality 数据，构建 one-row-per-ICU-stay 的 patient-level analysis dataset；最终分析队列 N=9,502。
- 完成 propensity score diagnostics、stabilized IPTW、weighted logistic regression 与 AIPW 分析，输出 cohort flow、Table 1/SMD 平衡性表、propensity overlap、love plot、粗死亡率比较和调整后的 risk difference。
- 比较粗分析与调整后结果：早期升压药组粗死亡率较高（32.9% vs. 25.2%），但 AIPW 调整后 risk difference 接近零（1.17 个百分点；95% CI: -0.52 至 2.87），体现 ICU 治疗策略分析中 confounding by indication 的影响。

技能栏建议 | 贴合 JD

模块	建议写法
Statistical & RWE Methods	Observational Study Design, Target Trial Emulation, Propensity Score, IPTW, AIPW, Logistic Regression, Table 1/SMD, TFL Outputs
Programming & Data	Python, SQL/DuckDB, R, Excel, PowerPoint; SAS/Stata 可写为 familiar/basic (仅在真实会用时写 proficient)
Healthcare Data	EHR/ICU Data, MIMIC-IV, Cohort Construction, Treatment Strategy Analysis, Real-World Evidence
Writing & Communication	Protocol/SAP-style documentation, study report drafting, bilingual scientific communication

面试 30 秒版本

我做过一个 MIMIC-IV ICU 的真实世界证据项目，问题是 sepsis-coded ICU 患者中 6 小时内早期升压药使用与院内死亡风险的关系。我没有只做粗死亡率比较，而是按 target trial emulation 的思路定义人群、time zero、grace period、暴露、结局和 confounders，然后用 SQL/DuckDB 和 Python 构建 patient-level analysis dataset，并完成 propensity score、IPTW、weighted logistic regression 和 AIPW。最终结果也体现了 RWE 项目中常见的 confounding by indication：粗死亡率早期升压药组更高，但调整后 risk difference 接近 null，因此解释时需要非常谨慎。

简历措辞风险控制

不要写	建议写
proved early vasopressor caused mortality change	estimated association / adjusted risk difference
demonstrated clinical benefit	assessed treatment strategy using observational RWE methods
published-level evidence	portfolio / project-level RWE analysis with reproducible extraction and TFL outputs